

Business Scenario for a University Management System

Introduction

Imagine you are a consultant at a software company specializing in developing university management systems. You have been asked to design a comprehensive system to manage students, professors, courses, departments, libraries, student clubs, and research projects. The goal is to improve data management efficiency and provide a user-friendly interface for both students and administrators.

System Requirements (Business Requirements)

Student Management

Each enrolled student has a unique ID.

A student may have a graduation project, and this project cannot exist independently of the student.

Some students may join student clubs, but it is not mandatory for all students.

Professors and Departments Management

Each professor is assigned to a specific department, and each department is supervised by one professor.

A professor may have an academic advisor within the same department.

Some professors supervise student projects, and every project must have at least one supervising professor.

Courses and Enrollment

A student can enroll in multiple courses each semester, and each course can be taken by multiple students.

The system must store students' grades for each course.

Each course has at least one responsible professor, but some courses may have multiple professors.

Library Management

The university has a central library with multiple books.

Each book registered in the system belongs to a specific library.

Students can borrow books, and the system must record the borrowing details, including the student who borrowed the book.

Student Clubs Management

Some students may join one or more student clubs, but not every student has to be part of a club.

Each club has a unique ID and name, and it may or may not have active members.

Research Projects and Collaboration

Some research projects require collaboration between a student, a professor, and a specific department.

Every project must be linked to a supervising professor and must belong to one of the university's departments.

Additional Student Information

Each student has basic details such as name, date of birth, and email.

Some students may have multiple phone numbers stored in the system.

The student's age is not stored directly but is calculated based on the date of birth.

Each student has a university ID card that contains a unique number and an issue date.

Project Challenges (Challenges)

How can we ensure that every department has at least one supervising professor?

How do we handle students who have more than one research project?

What is the best way to structure the relationship between students, projects, and professors efficiently?

How can we ensure accurate library book borrowing records?

How do we handle students who are not part of any club?

Final Objective of the Project

The goal is to design a robust and flexible system capable of managing all university-related data while ensuring that all entity relationships work correctly and meet the needs of both academic administration and students.

Your Task:

Analyze the scenario carefully and identify the main entities and their relationships.

Draw an ERD (Entity-Relationship Diagram) that accurately represents all the relationships in the project.

Convert the model into database tables correctly.

 Hint:

Pay attention to complex relationships such as ternary relationships, multivalued attributes, and relationships requiring an attributive entity.